HepatoFusion — Assignments #5 and #6

Task List, Milestones, Timeline, Effort Matrix, Work Breakdown Structure, and Risk Register

Project: HepatoFusion
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Partner: Cincinnati Children's Hospital
Courses: CS 5001 Senior Design I (Fall 2026) and CS 5002 Senior Design II (Spring 2027)
Document date: 28 October 2026
Canonical task file: `../Tasklist.md`

Each student submits this PDF separately. Both names appear on page 1. No PHI, MRNs, or slide files are in this document or in Git.

AI use. An AI tool helped format the tables. Task content, ownership, parked-vs-active scope, dates, hours, and risks are the team's, confirmed by both members on 28 October 2026.

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0. Scope control

D2 closed detailed design for the full pipeline: Identifier gate, Radiology intake, Genomics intake, Outcomes cleaner, Cohort case index, Modality integrator, Risk scorer, Review package, plus glue (Authentication, Notifications, Configuration, Logging). Implementation does **not** start every component at once.

Status	What	Why
Active through December 2026	Epic → Outcomes cleaner; eSlide Manager / PathPresenter → Cohort case index; Excel coding; IHC join spec	The spreadsheet and slide inventory are not model-ready.
Active in parallel, then Spring 2027	Existing pathology pipeline fine-tuning; Docker image for QuPath; Review package → PathPresenter	US-01: pathologist sees the tumor map in a tool already in the workflow. QuPath is the executable landing if PathPresenter overlay format (OQ-02) stays open.
Side project, does not block pathology	TMA grid study keys for the radiology collaborator	Starts the radiology data path without building Radiology intake this semester.
On ice	Radiology intake (beyond TMA keys), Genomics intake, Modality integrator, Risk scorer	D2 partial-score rule still requires pathology + cleaned outcomes + at least one of {radiology, molecular}. Ice is a date decision, not a design change.

IHC is not a new D0 box. It is additional pathology evidence recorded on the **Cohort case index**. Epic and eSlide Manager are external hospital systems, like the Outcomes spreadsheet and PathPresenter already drawn as dashed boxes.

Protected calendar: Thanksgiving 26–29 Nov 2026; exam week 6–12 Dec 2026; winter break 13 Dec 2026 – 10 Jan 2027; MLK Day 18 Jan 2027; spring break 15–21 Mar 2027; CS 5002 exam week 24–29 Apr 2027. Neither member is on a co-op rotation this academic year. Classes end 5 Dec 2026 (Fall) and 23 Apr 2027 (Spring).

0.1 Owners

Short name	Full name	Primary focus (`Team_Contract.md`)
Shivam	Shivam Sinay Kharangate	Pathology CV / multimodal ML; WSI pipeline; U-Net; clinical liaison
Nipun	Nipun Chandra	Data engineering / systems; ingestion; pipeline reliability; documentation

0.2 Course calendar (Week 1 14-week roadmap)

Weeks and graded components are taken from *CS 5001 Week 1: Course Overview & Team Formation* (Justin Zhan). Completed dates are this team's repository commits. Remaining CS 5001 Fridays run from this Week 8 submission through the last instructional week (classes end 5 Dec 2026).

Week	Topic (Week 1 slide 13)	Graded component (slide 12)	Date	Status
1	Course Overview & Teams	Team contract and team formation (5%)	29 Aug 2026	**Complete** — repo, roster
2	Problem Framing & Topics	Topic statement, bios, and reflective essay (10%)	9 Sep 2026	**Complete**
3	Advisors & Constraints	Advisor selection and constraints essay (5%)	17–18 Sep 2026	**Complete**
4	Requirements Elicitation	User Stories	28 Sep 2026	**Complete**
5	High-Level Architecture	High-level system design (10%)	2 Oct 2026	**Complete** — D0
6	Detailed Design Part 1	Detailed system design (15%, split Weeks 6–7)	8 Oct 2026	**Complete** — D1
7	Detailed Design Part 2 & Reviews	Detailed system design (15%, split Weeks 6–7)	14 Oct 2026	**Complete** — D2 + peer review
8	Project Management	Task timeline and effort matrix (10%)	30 Oct 2026	**This submission**
9	Requirements & Tech Spec Docs	Requirements and technical specification documents (15%)	6 Nov 2026	Remaining
10	Test & Validation Planning	Test and validation plan	13 Nov 2026	Remaining
11	Ethics & Responsible Use	Ethics / responsible-use write-up	20 Nov 2026	Remaining
12	Technical Communication	Technical communication artifact	25 Nov 2026	Remaining (short week; Thanksgiving 26–29 Nov)
13	Oral Presentations I	Oral presentation (15%)	2 Dec 2026	Remaining
14	Oral Presentations II & Wrap-Up	Design report, design notebook, and team assessment (15%)	4 Dec 2026	Remaining — report reprints Table 1 and Table 2

0.3 Effort method (PERT)

Expected hours = (O + 4M + P) / 6, rounded to a whole hour, confirmed by both owners. Calendar span is one or two weeks of owned work, plus **18% duration buffer** in Table 1, plus a **protected block** before each major milestone (Week 14 report, winter break, CS 5002 mid-demo, final presentation).

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1. Task List (Assignment #5)

Sixteen numbered tasks, one statement each, verb-first, one primary owner, mapped to D0–D2 names. Same text as `Tasklist.md`. Eight tasks per member.

T1. Obtain Epic clinical-outcome fields for the existing 102+ patient cohort and deliver them to the Outcomes cleaner on approved Cincinnati Children's storage. Owner: Shivam.

T2. Obtain the whole-slide inventory from eSlide Manager and PathPresenter and record study-ID membership in the Cohort case index. Owner: Nipun.

T3. Develop patient-by-patient cleanup of the clinical-outcomes spreadsheet so each agreed field is coded or numeric for the Outcomes cleaner. Owner: Nipun.

T4. Investigate immunohistochemistry (IHC) slide availability in eSlide Manager and PathPresenter and specify how IHC joins the Cohort case index. Owner: Shivam.

T5. Develop Cohort case index `case_link.json` records for cleaned pathology cases, including IHC when present and radiology as `not_linked`. Owner: Nipun.

T6. Investigate pathology U-Net outlier slides against Gold annotations in the existing pathology pipeline. Owner: Shivam.

T7. Refine the pathology U-Net on cleaned Gold labels after outlier review in the existing pathology pipeline. Owner: Shivam.

T8. Develop a Docker image of the pathology U-Net for QuPath inference on approved compute. Owner: Nipun.

T9. Research the PathPresenter overlay format (OQ-02) and develop Review package helper files for the existing tumor localization. Owner: Shivam.

T10. Test QuPath loading of the Dockerized U-Net on a holdout whole-slide image with no PHI in the course repository. Owner: Nipun.

T11. Obtain TMA-grid study keys on approved hospital systems for the radiology-collaborator handoff that will later feed Radiology intake. Owner: Shivam.

T12. Specify Identifier gate `identifier_rules` in `pipeline_config.yaml` for Epic and TMA exports before any file leaves the clinical system. Owner: Nipun.

T13. Develop the Outcomes cleaner ALG-01 run on the cleaned workbook and write model-ready `outcomes.json` plus the exclusion log. Owner: Nipun.

T14. Validate Review package open of the existing tumor localization inside PathPresenter with Dr. Patel. Owner: Shivam.

T15. Design Radiology intake registration for TMA-returned DICOM (parked until collaborators return studies). Owner: Nipun.

T16. Develop the Modality integrator `input_list.json` under the D2 partial-score rule (parked until pathology, cleaned outcomes, and at least one extra modality are linked). Owner: Shivam.

1.1 Module map

ID	D0–D2 module or named external system	Weeks	Requirement
T1	Outcomes cleaner	2	US-04, UC-02
T2	Cohort case index; existing pathology pipeline; PathPresenter	2	I6
T3	Outcomes cleaner	2	US-04, UC-02
T4	Cohort case index; existing pathology pipeline	2	I6
T5	Cohort case index	2	I6, I7, I9, I10
T6	Existing pathology pipeline	2	Project Description backbone
T7	Existing pathology pipeline	2	Project Description backbone
T8	QuPath; Review package	2	D0 D-08, OQ-05
T9	Review package; PathPresenter	2	US-01, OQ-02, I14

ID	D0–D2 module or named external system	Weeks	Requirement
T10	QuPath	2	OQ-05
T11	Radiology intake (side project)	2	US-03
T12	Identifier gate; Configuration	1	US-05, OQ-01, I17
T13	Outcomes cleaner	2	US-04, ALG-01
T14	Review package; PathPresenter	2	US-01, I15
T15	Radiology intake (ice)	2	US-03, I3, I7
T16	Modality integrator (ice)	1	US-02, I10–I12

Not given a Task List ID on purpose: Genomics intake, Risk scorer, Authentication, Logging, Notifications. They appear in the work breakdown structure. Authentication remains hospital/UC SSO. Logging is a by-product of T5 and T13. Notifications remain pull-only (`read_case_status`, I18). Genomics intake and Risk scorer stay ice with T15 and T16.

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2. Milestones List (Assignment #6)

A milestone is a dated, checkable outcome, not a task. Verification is what a reviewer can open or watch.

2.1 Completed course deliverables

ID	Milestone	Week 1 topic	Date	What it demonstrates	How verified	By whom	Status
M1	Repository and team roster	W1 Teams	29 Aug 2026	Team identity and contact path	Files in repo root	Instructor	**Complete**
M2	Project Description and biographies	W2 Topics	9 Sep 2026	HepatoFusion goal, 102+ patient cohort, pathology U-Net metrics, advisor through April 2027	`Project-Description.md`; bios	Instructor	**Complete**
M3	Team Contract	W1 / W3	17 Sep 2026	Two-person roles, PHI rule, meeting cadence	`Team_Contract.md` signed SK / NC / KP	Advisor + instructor	**Complete**
M4	Constraints Essay	W3 Advisors & Constraints	18 Sep 2026	PHI, compute, research-prototype, no FDA claim	`Constraints_Essay_sinayksp.md`	Instructor	**Complete**
M5	User Stories and Use Cases	W4 Requirements Elicitation	28 Sep 2026	US-01–US-05, UC-01–UC-02	`User_Stories.md`	Instructor	**Complete**
M6	D0 High-Level Design	W5 High-Level Architecture	2 Oct 2026	Eight built components; PathPresenter first; QuPath alternate	`Design_Diagrams/D0_High_Level_Design.md`	Instructor	**Complete**
M7	D1 Detailed Design	W6 Detailed Design Part 1	8 Oct 2026	Cohort case index; Outcomes cleaner ALG-01	`Design_Diagrams/D1_Detailed_Design.md`	Instructor	**Complete**

ID	Milestone	Week 1 topic	Date	What it demonstrates	How verified	By whom	Status
M8	Peer design review	W7 Reviews	9 Oct 2026	External critique of D1 before D2	Review session; notes in `Peer_Review_Notes.md`	Peer teams + instructor	**Complete**
M9	D2 Detailed Design	W7 Detailed Design Part 2	14 Oct 2026	Remaining components, error paths, partial-score rule, OQ-01–OQ-05	`Design_Diagrams/D2_Detailed_Design.md`	Instructor	**Complete**

2.2 Remaining CS 5001 deliverables through Week 14

ID	Milestone	Week 1 topic	Date	What it demonstrates	How verified	By whom	Status
M10	Task List + implementation plan	W8 Project Management	30 Oct 2026	Sixteen owned tasks, dates, 50/50 hours, risks	`Tasklist.md` + this PDF	Instructor; advisor	This submission
M11	Requirements and technical specification documents	W9 Requirements & Tech Spec Docs	6 Nov 2026	Specs aligned with D2 and the pathology-first scope, including IHC as pathology evidence	Spec PDF/MD in repo	Instructor	Planned
M12	Test and validation plan	W10 Test & Validation Planning	13 Nov 2026	How T10, T13, and T14 are checked with no PHI in Git	Test-plan PDF/MD in repo	Instructor	Planned
M13	Ethics and responsible-use write-up	W11 Ethics & Responsible Use	20 Nov 2026	PHI, research-prototype disclaimer, no diagnostic claim	Canvas PDF	Instructor	Planned
M14	Technical communication artifact	W12 Technical Communication	25 Nov 2026	Audience-ready write-up of the pathology-first plan	Canvas PDF	Instructor	Planned (before Thanksgiving)
M15	Oral Presentations I	W13 Oral Presentations I	2 Dec 2026	Pathology-first plan, ice list, TMA side project, QuPath/PathPresenter path	Live or recorded talk	Instructor; advisor invited	Planned
M16	Oral Presentations II, design report, notebook, team assessment	W14 Wrap-Up	4 Dec 2026	Design report reprints Table 1 and Table 2 with actual vs. plan; notebook; team assessment	Report PDF; notebook; Canvas	Instructor; Dr. Patel	Planned
M17	Fall data-cleanup checkpoint	Internal	4 Dec 2026	Epic fields landed; eSlide inventory in the Cohort case index; first-pass Excel coded; TMA keys handed off on the hospital side; Identifier gate rules written	Inventory + cleanup counts on approved storage; no PHI in Git	Shivam (clinical); Nipun (files); Dr. Patel (spot-check)	Internal, reported in M16

2.3 Internal and CS 5002 milestones (every one to three weeks through April 2027)

Winter break is empty on purpose. CS 5002 dates follow the Week 1 “Build & Deliver” semester: implement, test, iterate, handoff documentation, final product, report, and public presentation.

ID	Milestone	Date	What it demonstrates	How verified	By whom	Status
M18	Identifier gate rules frozen	6 Nov 2026	`identifier_rules` version in `pipeline_config.yaml`	Config on approved share; no identifier values	Nipun; Shivam liaison	Planned
M19	Epic + eSlide extracts exist	13 Nov 2026	T1 and T2 produced files on approved storage	Study-ID list (no names)	Shivam; Nipun	Planned

ID	Milestone	Date	What it demonstrates	How verified	By whom	Status
M20	TMA keys delivered to radiology collaborator	20 Nov 2026	Side project moved off the search list	Collaborator acknowledgement on the hospital system, not Git	Shivam	Planned
M21	IHC join specified	21 Nov 2026	Written rule: how an IHC slide attaches to `case_link.json`	Spec paragraph in repo (no slide files)	Shivam; Dr. Patel	Planned
M22	Thanksgiving / exam / winter protected block	26 Nov 2026 – 10 Jan 2027	No task completion dates inside this window	Table 1 empty in this range	Both	Planned
M23	CS 5002 plan freeze	15 Jan 2027	Spring owners unchanged unless M16 said otherwise	Advisor meeting note	Dr. Patel	Planned
M24	ALG-01 first clean run	23 Jan 2027	Model-ready `outcomes.json` + exclusion log with three counts	Summary dict from ALG-01; no PHI	Nipun; Shivam checks join keys	Planned
M25	Outlier list closed	4 Dec 2026 (inventory) / 23 Jan 2027 (labels)	Gold slides the U-Net fails, listed by `study_id`	List on approved share	Shivam	Planned
M26	U-Net refinement checkpoint	30 Jan 2027	Retrain on cleaned Gold; holdout Dice/AUROC vs. Project Description baseline (0.96 mean slide Dice, 0.99 tumor AUROC). No claim of improvement until the number exists	Metrics table; weights on approved compute	Shivam; Dr. Patel	Planned
M27	QuPath Docker image builds	30 Jan 2027	Container runs the current U-Net on approved compute	`docker build` log (no WSI in Git)	Nipun	Planned
M28	CS 5002 progress check 1	5 Feb 2027	Actual vs. Table 1 for T7, T8, T13	Short PDF; tables reprinted	Instructor; advisor	Planned
M29	QuPath holdout inference	13 Feb 2027	Dockerized model opens one holdout WSI in QuPath	Deidentified screenshot on hospital/UC machine	Nipun; Shivam	Planned
M30	Review package overlay prototype	13 Feb 2027	Helper files from T9; PathPresenter import tried (OQ-02)	`review_package.json` + viewer_open status	Shivam	Planned
M31	CS 5002 mid-implementation demo	26 Feb 2027	Live QuPath Docker inference and/or PathPresenter helper. Disclaimer spoken: research prototype, not a diagnostic device	Demo to advisor and instructor	Both; Dr. Patel	Planned
M32	PathPresenter US-01 validation	12 Mar 2027	Dr. Patel opens a deidentified case and sees the existing tumor map as a helper	Advisor written OK	Dr. Patel (verifier); Shivam	Planned
M33	CS 5002 progress check 2	12 Mar 2027	Pre-break status; T14 result; ice/thaw decision for T15	PDF	Instructor	Planned
M34	Spring break protected block	15–21 Mar 2027	No completions	Table 1	Both	Planned
M35	CS 5002 progress check 3	2 Apr 2027	Effort-matrix actuals	PDF	Instructor	Planned
M36	Ice/thaw gate for radiology	3 Apr 2027	If TMA-linked DICOM arrived, T15 is real; if not, T15 is the parked schema only and T16 does not claim a score	Written gate note	Nipun; Shivam	Planned
M37	Integrator dry run or explicit skip	10 Apr 2027	`input_list.json` for one case, or a written skip (`insufficient_modalities`)	File or skip note	Shivam	Planned

ID	Milestone	Date	What it demonstrates	How verified	By whom	Status
M38	Final presentation / demo	16 Apr 2027	Pathology cleanup, U-Net refinement, QuPath Docker, PathPresenter helper; ice items labeled ice	Talk + demo	Instructor; advisor	Planned
M39	CS 5002 final report	23 Apr 2027	Goals vs. achievements; reprinted Table 1–2 with actuals	Final report PDF	Instructor; advisor	Planned

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3. Table 1 — Timeline (Assignment #6)

One row per Task List item, then one row per milestone completion date. Buffer is explicit: **18% added to each task's working duration**, plus protected blocks before M16, M22, M31, and M38.

Critical path (fall checkpoint M17): T1 → T3 → M17, with T12 on the PHI gate before any export. Parallel: T2 → T4 → T5; T2 → T6; T12 → T11 → M20.

Critical path (CS 5002 demo M31): T6 → T7 → T9 and T8 → T10. T14 is the advisor check after that.

Critical path (optional fusion M37): T11 → M36 → T15 → T16, and T13 → T16. This path is allowed to stop at M36 without failing the senior-design prototype.

3.1 Task rows

ID	Statement	Owner	Start	Finish (+18%)	Dependencies	Buffer / notes
T12	Specify Identifier gate `identifier_rules` in `pipeline_config.yaml`	Nipun	2026-10-28	2026-11-06	None	1 week + 1d. PHI gate before Epic/TMA leave the clinical system.
T1	Obtain Epic fields for Outcomes cleaner	Shivam	2026-10-28	2026-11-13	None	2 weeks + 2d. Hospital access risk R1.
T2	Obtain eSlide Manager / PathPresenter inventory for Cohort case index	Nipun	2026-10-28	2026-11-10	None	2 weeks + 2d. Parallel with T1.
T11	Obtain TMA-grid study keys for Radiology intake handoff	Shivam	2026-11-02	2026-11-20	T12	2 weeks + 2d. Side project. Keys stay on hospital systems.
T3	Develop patient-by-patient Excel cleanup for Outcomes cleaner	Nipun	2026-11-09	2026-11-24	T1	2 weeks + 2d. Stops before Thanksgiving. Remaining free-text is R2.
T4	Investigate IHC join into Cohort case index	Shivam	2026-11-09	2026-11-21	T2	2 weeks + 2d.
T5	Develop `case_link.json` records	Nipun	2026-11-16	2026-12-04	T2, T4; uses T3 as it lands	2 weeks of work; Thanksgiving pause 26–29 Nov. Radiology slot = `not_linked`.
T6	Investigate U-Net outliers vs Gold	Shivam	2026-11-16	2026-12-04	T2	Thanksgiving pause. List by `study_id` only.
T13	Develop Outcomes cleaner ALG-01 run	Nipun	2027-01-12	2027-01-23	T3, T12	2 weeks + 2d after winter. MLK 18 Jan is a day off.
T7	Refine pathology U-Net	Shivam	2027-01-12	2027-01-30	T6	2 weeks + 3d. Train only on approved compute.
T8	Develop Docker image for QuPath	Nipun	2027-01-12	2027-01-30	None (current weights; swap T7 weights when ready)	2 weeks + 3d. Parallel with T7 and T13.

ID	Statement	Owner	Start	Finish (+18%)	Dependencies	Buffer / notes
T9	Research PathPresenter format; develop Review package	Shivam	2027-02-01	2027-02-13	T7 (maps); OQ-02 conversation already open	2 weeks + 2d. If OQ-02 is still closed, T9 still writes GeoJSON + helper JSON (D2).
T10	Test QuPath Docker on holdout WSI	Nipun	2027-02-01	2027-02-13	T8	2 weeks + 2d. Protected 14–25 Feb before M31.
T14	Validate PathPresenter helper with Dr. Patel	Shivam	2027-03-01	2027-03-12	T9	2 weeks. Protected 13 Mar, then spring break.
T15	Design Radiology intake for TMA-returned DICOM	Nipun	2027-03-23	2027-04-03	T11; **ice unless M36 thaws**	2 weeks + 2d. If no DICOM, deliverable is the parked schema.
T16	Develop Modality integrator `input_list.json`	Shivam	2027-04-06	2027-04-10	T5, T13, T15	1 week. Ice unless M36 thaws. Protected 11–15 Apr before M38.

No task completes during 6–12 Dec 2026, 13 Dec 2026 – 10 Jan 2027, 15–21 Mar 2027, or 24–29 Apr 2027.

3.2 Milestone rows (planned completion dates)

ID	Milestone	Planned completion	Depends on
M1–M9	Completed CS 5001 artifacts	See §2.1	—
M10	This plan submitted (W8)	2026-10-30	—
M18	Identifier gate rules frozen	2026-11-06	T12
M11	Requirements and tech spec docs (W9)	2026-11-06	M10
M12	Test and validation plan (W10)	2026-11-13	M10
M19	Epic + eSlide extracts exist	2026-11-13	T1, T2
M20	TMA keys delivered	2026-11-20	T11
M13	Ethics write-up (W11)	2026-11-20	—
M21	IHC join specified	2026-11-21	T4
M14	Technical communication (W12)	2026-11-25	—
Protected	Thanksgiving	2026-11-26 to 2026-11-29	—
M15	Oral Presentations I (W13)	2026-12-02	M10
M16	Oral Presentations II + design report (W14)	2026-12-04	Table 1–2 reprint; T3 done
M17	Fall data-cleanup checkpoint	2026-12-04	T1, T2, T3, T5, T11, T12
M25	Outlier list closed (inventory pass)	2026-12-04	T6
Protected	Exams + winter break	2026-12-06 to 2027-01-10	—
M23	CS 5002 plan freeze	2027-01-15	M16

ID	Milestone	Planned completion	Depends on
M24	ALG-01 first clean run	2027-01-23	T13
M25b	Outlier list closed (label pass)	2027-01-23	T6 follow-up
M26	U-Net refinement checkpoint	2027-01-30	T7
M27	QuPath Docker builds	2027-01-30	T8
M28	CS 5002 progress check 1	2027-02-05	M26, M27
M29	QuPath holdout inference	2027-02-13	T10
M30	Review package overlay prototype	2027-02-13	T9
Protected	Pre-demo buffer	2027-02-14 to 2027-02-25	T10, T9
M31	Mid-implementation demo	2027-02-26	M29 or M30
M32	PathPresenter US-01 validation	2027-03-12	T14
M33	CS 5002 progress check 2	2027-03-12	T14
Protected	Spring break	2027-03-15 to 2027-03-21	—
M35	CS 5002 progress check 3	2027-04-02	Effort actuals
M36	Ice/thaw gate	2027-04-03	T15
M37	Integrator dry run or skip	2027-04-10	T16
Protected	Pre-final buffer	2027-04-11 to 2027-04-15	—
M38	Final presentation	2027-04-16	M29, M32; T15/T16 optional
M39	Final report	2027-04-23	All active tasks

4. Table 2 — Effort Matrix (Assignment #6)

Hours are PERT expected hours from §0.3, confirmed by both members. The **primary owner is marked *** and always has the larger share of that row. Secondary hours are review, pairing, or hospital-access support, not a second owner.

4.1 Three-point estimates

ID	O	M	P	Expected (O+4M+P)/6
T1	10	16	22	16
T2	8	12	16	12
T3	12	19	32	20

ID	O	M	P	Expected (O+4M+P)/6
T4	8	12	16	12
T5	8	14	20	14
T6	10	16	22	16
T7	12	18	36	20
T8	10	16	22	16
T9	10	16	22	16
T10	8	12	16	12
T11	6	10	14	10
T12	6	10	14	10
T13	8	14	20	14
T14	6	10	14	10
T15	8	14	20	14
T16	10	16	22	16
Total				**228**

4.2 Matrix (hours)

Task	Module	Shivam	Nipun	Row total
T1 Obtain Epic fields	Outcomes cleaner	**13 ***	3	16
T2 Obtain eSlide / PathPresenter inventory	Cohort case index	3	**9 ***	12
T3 Develop Excel cleanup	Outcomes cleaner	3	**17 ***	20
T4 Investigate IHC join	Cohort case index	**9 ***	3	12
T5 Develop `case_link.json`	Cohort case index	3	**11 ***	14
T6 Investigate U-Net outliers	Existing pathology pipeline	**13 ***	3	16
T7 Refine pathology U-Net	Existing pathology pipeline	**16 ***	4	20
T8 Develop Docker image for QuPath	QuPath / Review package	3	**13 ***	16
T9 Research overlay; develop Review package	Review package / PathPresenter	**12 ***	4	16
T10 Test QuPath Docker inference	QuPath	3	**9 ***	12
T11 Obtain TMA-grid keys	Radiology intake (side)	**8 ***	2	10

Task	Module	Shivam	Nipun	Row total
T12 Specify Identifier gate rules	Identifier gate / Configuration	2	**8 ***	10
T13 Develop ALG-01 run	Outcomes cleaner	3	**11 ***	14
T14 Validate PathPresenter helper	Review package / PathPresenter	**8 ***	2	10
T15 Design Radiology intake (ice)	Radiology intake	3	**11 ***	14
T16 Develop Modality integrator (ice)	Modality integrator	**12 ***	4	16
Column total (hours)		**114**	**114**	**228**
Column share		**50%**	**50%**	**100%**

Primary-owner hours (the * cells): Shivam 91, Nipun 89. Support hours are the remainder.

4.3 Balance

Columns are **114 / 114 (50% / 50%)**. No reassignment is required at plan time.

If actuals drift:

1. T3 is the overrun candidate (patient-by-patient Excel, pessimistic 32 h). If Nipun exceeds 17 h on T3 before 24 Nov, Shivam takes the next batch of agreed-field coding under Nipun's protocol. Primary owner of T3 stays Nipun.
2. If T7 GPU time slips, T9 still proceeds with current maps. US-01 does not wait on a better Dice.
3. T15 and T16 are ice. If Spring load is high, drop T16 first (M37 becomes an explicit skip).
4. Week 14 (M16) and CS 5002 progress checks reprint this table with actual hours. The syllabus uses the effort matrix when adjusting individual grades from the team grade.

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5. Work breakdown structure

Two levels: phases, then Task List items. Every T-ID appears exactly once.

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HepatoFusion
│
├── 1. Course design (complete; no new Task List IDs)
│      ├── M1–M4  roster, Project Description, Team Contract, Constraints
│      ├── M5     User Stories
│      └── M6–M9  D0, D1, peer review, D2
│
├── 2. Data access and cleanup  [main effort through 4 Dec 2026]
│      ├── T12  Identifier gate + Configuration (`pipeline_config.yaml`)
│      ├── T1   Epic → Outcomes cleaner (obtain)
│      ├── T2   eSlide Manager / PathPresenter → Cohort case index (obtain)
│      ├── T3   Excel patient-by-patient coding (develop)
│      ├── T4   IHC join spec (investigate / specify)
│      ├── T5   `case_link.json` writes, radiology = not_linked (develop)
│      └── T13  ALG-01 model-ready table (develop)   [runs after winter]
│
├── 3. Existing pathology pipeline
│      ├── T6   Outlier investigation vs Gold (investigate)
│      └── T7   U-Net refine on cleaned Gold (refine)
│
├── 4. Viewer integration (US-01)
│      ├── T8   Docker image of U-Net for QuPath (develop)
│      ├── T10  QuPath holdout test (test)
│      ├── T9   Review package + PathPresenter overlay (research / develop)
│      └── T14  PathPresenter validation with Dr. Patel (validate)
│
├── 5. TMA side project (Radiology intake, not on the pathology critical path)
│      └── T11  Study keys for radiology collaborator (obtain)
│
├── 6. On ice until M36
│      ├── T15  Radiology intake for returned DICOM (design)
│      ├── T16  Modality integrator under D2 partial-score rule (develop)
│      ├── (no T-ID) Genomics intake
│      └── (no T-ID) Risk scorer   [interface exists in D2; model is OQ-04]
│
└── 7. Glue already decided in D2 (no extra Task List IDs)
       ├── Authentication  – SSO / folder ACL only
       ├── Logging         – emitted by T5 and T13 writes (I16)
       └── Notifications   – none pushed; I18 status read only
```

---

6. Risk register

Seven risks across technical, team, and external. High-impact schedule risks are visible in Table 1 (buffers, ice, protected blocks).

ID	Category	Risk	Likelihood	Impact	Strategy	Mitigation	Owner	Trigger
R1	External	Epic or eSlide Manager access slips, so T1/T2 miss 13 Nov	Medium	High (blocks T3, T5, M17)	Reduce	Start T1/T2 on 28 Oct; 18% buffer; if no extract by **14 Nov**, Shivam escalates to Dr. Patel the same day and T3 runs on fields already in the spreadsheet	Shivam	No approved extract by 14 Nov 2026
R2	Technical	Outcomes spreadsheet still has free-text after T3	High	High (T13 and any later score stay blocked; UC-01 E1)	Reduce	T3 stops 24 Nov with a counted leftover list; T13 in January processes only ready rows; `field_not_coded` is success, not failure	Nipun	Kept-row count < 70% of source at M17
R3	External	PathPresenter overlay format (OQ-02) still unknown in February	Medium	High (US-01 PathPresenter path)	Reduce	T8/T10 make QuPath the executable viewer; T9 still writes D2 GeoJSON + helper JSON; M31 can demo QuPath if PathPresenter refuses	Shivam	No written import format from CCHMC IT by 1 Feb 2027
R4	Technical	U-Net refinement (T7) does not beat the Project Description baseline	Medium	Medium (demo still uses existing maps)	Accept + reduce	T9/T14 use the current localization if holdout Dice does not move	Shivam	Holdout mean slide Dice < 0.96 after T7
R5	External	Radiology collaborator does not return TMA-linked DICOM	High	High for fusion; **low for the pathology prototype**	Accept	T15/T16 are ice. M36 is a gate. T11 still delivers keys	Shivam	No DICOM on the approved share by 23 Mar 2027
R6	Team	Two-person team; exam week, winter break, or one illness hits T3/T7	Medium	High	Reduce	Protected blocks in Table 1; 48-hour blocker rule from the Team Contract; T3 protocol written so Shivam can code rows without becoming owner	Nipun (calendar), both (coverage)	Missed standup, or a task still open 3 days after its Table 1 end
R7	Technical / external	PHI (MRN, name) lands in Git or in a Docker context	Low	High (data-use breach; US-05)	Avoid	T12 before T1/T11 exports; Identifier gate blocks rather than guesses; no WSI, DICOM, or identified Excel in this repo	Nipun	MRN, name, DOB, or address in a commit, log, or `Dockerfile`

R1, R2, R3, R5, and R6 are the schedule risks. They show up as: early T1/T2, leftover-row honesty at M17, QuPath parallel to PathPresenter, ice on T15/T16, and the empty winter/exam/break cells in Table 1.

7. Consistency pass

Check	Result
Sixteen tasks, 10–25 range	Yes
Verb-first, one statement, one owner	Yes
Tasks per member	Eight each
Shivam IDs	T1, T4, T6, T7, T9, T11, T14, T16
Nipun IDs	T2, T3, T5, T8, T10, T12, T13, T15
Same ID/name/owner in Task List, WBS, Table 1, Table 2	Yes

Check	Result
D0–D2 names	Outcomes cleaner, Cohort case index, existing pathology pipeline, Identifier gate, Configuration, Review package, PathPresenter, QuPath, Radiology intake, Modality integrator
Primary owner has the largest share on every row	Yes
Row and column totals	228 h; **114 / 114 (50% / 50%)**
Ice visible in Table 1	T15, T16 start after M36
Remaining CS 5001 weeks match Week 1 roadmap	W8–W14 as in §0.2
`Tasklist.md` at repo root	Yes

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8. Work starting 28 October 2026

Shivam: T1 Epic field list with Dr. Patel; start T11 TMA key search on approved systems only; keep the OQ-02 IT thread open for T9.

Nipun: T12 `identifier_rules` stub in `pipeline_config.yaml`; T2 eSlide Manager / PathPresenter inventory by `study_id`; create the approved-share folders named in D2 Appendix B, empty of PHI.